Lesson plan · for the teacher

Quick rules and their traps

Optimization · Level 1 · Lesson 3

Time
40 min

Materials
Nine name cards for the “villages”; A ball of string to lay out the route; Board and chalk for the step counts

Learning goals

- Make a plan fast with a quick rule.
- Improve the plan with small swaps.
- Ask who pays for a rule, not only the total.

Activity: A human bus route

1. Warm-up (5 min): Ask pupils how a bus or minibus driver might choose the order of stops. Collect ideas and point out “nearest next” as a quick rule.
2. Teach (8 min): Draw five dots and the school on the board. Build a route nearest first, then show one crossing and how swapping the order between the two roads removes it.
3. Bus round (15 min): Nine pupils stand around the room as villages, with name cards; the door is the school. The “bus” walks nearest first, laying string behind and counting steps. Write the total on the board.
4. Swap (7 min): The class finds places where the string crosses. Swap the order of the villages between them, re-lay the string and count again. Keep the swap only if the count went down.
5. Wrap-up (5 min): Which village was picked up first and rode the longest? Would starting at the far end be fairer? Agree on the best round the class found.

Spread the “villages” unevenly, with a cluster on one side and two far away, so nearest first leaves long legs at the end. A ball of string makes crossings easy to see.

Discussion

- Why are the last legs of a nearest-first round so long?
- Why not just try every possible round?
- Who pays for the nearest-first rule, and how could we share that fairly?

Exit ticket

- Name one quick rule we use at school, one time it could fail, and how we could check it.

Try it in class tomorrow
Make a human bus route: nine pupils stand around the room as villages, and a “bus” pupil walks nearest first from the door, counting steps. Then the class looks for paths that cross, swaps them, and counts again. Ask: who rode the longest, and is that fair?

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Three swaps took the bus round from 34.6 km to 28.6 km. What did the swaps do?**

- ◯ They made the bus drive faster
- ◯ They dropped a village from the round
- ◯ They untangled crossing roads, so the round got shorter

2 **With 20 villages, a computer can check every possible bus round in a few minutes.**

- ◯ True
- ◯ False

3 **When is a quick rule a good choice? Choose all that apply.**

- ☐ You need a plan in minutes, not days
- ☐ You will check it and improve it afterwards
- ☐ Nobody will ever look at the result
- ☐ The problem is too big to try every plan
- ☐ You must be sure the plan is the very best

4 **Match each quick rule to the school problem it plans.**

Nearest village next	• • • •	Loading the supply truck
Most points per kilo first	• • • •	Planning exam week
Most votes first	• • • •	The morning bus round
The exam with the most clashes gets a slot first	• • • •	Choosing the class outings

5 **Complete the sentence.**

Nearest neighbour builds a round _____ , then 2-opt swaps remove _____ and keep only changes that make it _____ .

Word bank: villages · shorter · fast · longer · crossings

Answers · for the teacher

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1 **They untangled crossing roads, so the round got shorter**

A 2-opt swap takes out two roads and joins the stretch between them the other way round.

2 **False**

Through 20 villages there are more than a billion billion rounds.

3 **The problem is too big to try every plan; You need a plan in minutes, not days; You will check it and improve it afterwards**

Quick rules shine when the problem is big or time is short, especially followed by checking and improving.

4 **Nearest village next — The morning bus round**
Most points per kilo first — Loading the supply truck
Most votes first — Choosing the class outings
The exam with the most clashes gets a slot first — Planning exam week

Every kind of school decision has its own quick rules.

5 **Nearest neighbour builds a round fast, then 2-opt swaps remove crossings and keep only changes that make it shorter.**

A quick rule gives a starting plan; improvement keeps every village but removes crossings, accepting a change only when the round gets shorter.

For you to think about

Which quick rule does your school use without thinking: seniority, first come first served, alphabetical order? When might it miss the best plan, and who pays for it?